Name: _____

TOPIC TEST 2

Chapter 10: Statistical inference

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 8 multiple-choice questions (8 marks).
- Part B: 9 short-answer questions (30 marks).
- Part C: 2 analysis questions (22 marks).

Part A

Multiple-choice questions

8 multiple-choice questions
1 mark each: 8 marks
Circle the correct answer.

1 What is the mean and standard deviation of the results below?

34, 72, 17, 78, 71, 37, 60, 78, 47, 45

A 59.9, 20.0

B 52.9, 21.1

C 53.9, 20.0

D 54.9, 20.0

E 53.9, 21.1

[1 mark]

2 A sample of 45 items has a mean of 15 and a standard deviation of 5. What is the standard deviation of the sampling mean?

A 0.75

B 1.73

C 1.29

D 2.23

E 3.87

[1 mark]

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3 The sampling distribution of the mean mass of oranges was found to have a mean of 306 g with a standard deviation of 8 g for samples of 30. What would the standard deviation of individual oranges be?

A 10 g

B 1.5 g

C 140 g

D 44 g

E 85 g

[1 mark]

4 The average mass of bags of potatoes is 50.2 kg with a standard deviation of 2.5 kg. What is the probability that the next 5 bags will have an average mass less than 49 kg?

A 0.3156

B 0.8229

C 0.3237

D 0.1416

E 0.8584

[1 mark]

5 The average total of sales at a butcher's shop is $35.60, with a standard deviation of $29.40. What is the probability that the next 10 sales have a total between $400 and $500?

A 0.3121

B 0.1288

C 0.2573

D 0.6820

E 0.4405

[1 mark]

6 What is the 99% confidence interval for a normal variable with mean 126 and standard deviation 18?

A [90.7, 161.3]

B [108.2, 143.8]

C [72, 180]

D [79.6, 172.4]

E [106.9, 142.6]

[1 mark]

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7 What is the confidence level for the interval 48–62 for a normal variable with mean 55 and standard deviation 3?

A 99%

B 40%

C 95%

D 98%

E 90%

[1 mark]

8 What is the 90% confidence interval for the population mean if the mean of a sample of 60 is 62 with a standard deviation of 12?

A [33.5, 90.5]

B [59.5, 64.5]

C [51.2, 72.8]

D [59, 65]

E [46.4, 77.6]

[1 mark]

Part B

Short-answer questions

9 short-answer questions
30 marks
Show your working where appropriate.

1 **a** A normal die with faces numbered 1 to 6 is tossed 6 times and the average is calculated. This is done 1200 times. What is the graph of the distribution likely to look like?

b Instead of graphing the results directly, the averages of the results in groups of 30 sets of 6 are found. What is the graph of these averages likely to look like?

[8 marks]

2 **I** A sample of 40 is chosen from a binomial distribution with $p = 0.4$ and $n = 30$ and a graph of the sample is drawn.

II 30 samples of 40 are chosen from a binomial distribution with $p = 0.4$ and $n = 30$ and a graph of the means of each sample is drawn.

III 30 samples of 100 are chosen from a binomial distribution with $p = 0.4$ and $n = 30$ and a graph of the means of the samples is drawn.

How will the graphs from parts **I**, **II** and **III** be likely to compare?

[6 marks]

3 What is the mean and standard deviation of a sampling distribution of a binomial distribution with $p = 0.65$ and $n = 30$ for samples of 40?

[3 marks]

4 What is the mean and standard deviation of a sampling distribution of a uniform distribution on the interval [10, 24] for samples of 60?

[3 marks]

5 A pallet with 60 cartons of apples weighs 728 kg. The pallet weighs 15 kg and the cartons are 450 g each. Each carton has 75 apples and the standard deviation of the masses of the cartons is 0.095 kg. What is the individual average and standard deviation of the weights of the apples?

[3 marks]

6 What is the 96% confidence interval for the standard normal variable Z?

[2 marks]

7 What is the 95% confidence interval for the normal variable X with mean 58 and variance 20?

[3 marks]

8 What is the confidence level for the interval [32, 46] for a normal variable with a mean of 37 and a standard deviation of 3?

[3 marks]

9 What is the 95% confidence interval for the population mean if a sample of 40 had a mean of 43 and a variance of 25?

[3 marks]

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Part C

Analysis questions

2 analysis questions
22 marks
Show your working where appropriate.

1 From a survey of 500 people, it was reported that the 94% confidence interval for their average daily sleeping time was 7.2 hours to 7.7 hours. What is the probability that, if you asked another 8 people, they would spend average of less than 8 hours a day asleep?

[8 marks]

2 The populations of Australian states and territories are:

New South Wales:	7.5 million
Victoria:	5.8 million
Queensland:	4.7 million
Western Australia:	2.5 million
South Australia:	1.7 million
Tasmania:	0.5 million
Australian Capital Territory:	0.4 million
Northern Territory:	0.3 million

A national survey of the amounts Australians spent on entertainment found it was $380 a month with a margin of error of 4%. The survey was taken by phoning a stratified sample of 800 people on the electoral rolls from the capital cities and asking the 'head of the household'. The calculated variances for each capital city were approximately the same.

a What does the stated margin of error mean?

b How many from each capital were interviewed?

c What are the margins of error for the individual states and territories?

d State any problems with the survey.

[11 marks]

[Total marks: 60]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 $\Sigma x = 539$, so mean $= 53.9$, $\Sigma x^2 = 33041$ so $\sigma = \sqrt{\frac{33041}{10} - 53.9^2} = 19.972\ldots$

$\therefore$ **C** **[1 mark]**

2 $\frac{5}{\sqrt{45}} = 0.745\ldots$

$\therefore$ **A** **[1 mark]**

3 $8 \times \sqrt{30} = 43.81\ldots$

$\therefore$ **D** **[1 mark]**

4 $\text{normCDf}\left(-\infty, 49, 50.2, \frac{2.5}{\sqrt{5}}\right) = 0.141\,565\ldots$

$\therefore$ **D** **[1 mark]**

5 $\text{normCDf}\left(40, 50, 35.6, \frac{29.4}{10}\right) = 0.257\,305\ldots$

$\therefore$ **C** **[1 mark]**

6 $2.576 \times 18 = 46.368$, $126 - 46.368 = 79.632$, $126 + 46.368 = 172.368$

$\therefore$ **D** **[1 mark]**

7 $\text{normCDf}(48, 62, 55, 3) = 0.980\ldots$

$\therefore$ **D** **[1 mark]**

8 $1.645 \times \frac{12}{\sqrt{60}} = 2.548\ldots$, $62 \pm 2.548\ldots \approx [59.5, 64.5]$

$\therefore$ **B** **[1 mark]**

Part B – Short-answer questions

1 **a** It is symmetrical around the peak of 3.5, gradually tapering off towards 1 and 6. The standard deviation is about 0.916. **[2 marks]**

b It is symmetrical around the peak of 3.5, but tapers off quickly to about 2 and 5. The standard deviation is about 0.17. It is more concentrated than **a**. **[2 marks]**

[4 marks altogether]

2 **I** The graph is skewed to the right, with a mean of 12 and a standard deviation of about 2.68. **[2 marks]**

II The graph is less skewed than **I**, with the same mean but a standard deviation of only about 0.42. It is much more concentrated than **I**. **[2 marks]**

III The graph is almost symmetrical, with the same mean, 12, but a standard deviation of only about 0.27 and is very concentrated in the centre. **[2 marks]**

[6 marks altogether]

3 Mean $= 0.65 \times 30 = 19.5$, **[1 mark]**

$\text{SD} = \sqrt{\frac{30 \times 0.65 \times 0.35}{40}} \approx 0.413$ **[2 marks]**

[3 marks altogether]

4 Mean $= 0.5(10 + 24) = 17$, **[1 mark]**

$$\text{SD} = \frac{24 - 10}{\sqrt{12} \times \sqrt{60}} \approx 0.521$$ **[2 marks]**

[3 marks altogether]

5 Average mass of 1 apple $= (713 \text{ kg} \div 60 - 0.45) \div 75 \approx 152.4 \text{ g}$ **[1 mark]**

Standard deviation of average mass of 75 apples $= 0.095 \div 60 = 1.583\ldots \text{ g}$ **[1 mark]**

Standard deviation of an individual apple $= 1.583\ldots \times \sqrt{75} \approx 13.7 \text{ g}$ **[1 mark]**

[4 marks altogether]

6 invNormCDf(0.02,0,1) ≈ -2.054 **[1 mark]**

96% confidence interval is $-2.054, 2.054]$ **[1 mark]**

[2 marks altogether]

7 $58 - 1.96 \times \sqrt{20} \approx 49.2$ **[1 mark]**

$58 + 1.96 \times \sqrt{20} \approx 66.8$ **[1 mark]**

90% confidence interval is [49.2, 66.8] **[1 mark]**

[3 marks altogether]

8 normCDf(32, 46, 37, 3) $= 0.9508\ldots$ **[2 marks]**

The confidence level is about 95.1% **[1 mark]**

[3 marks altogether]

9 $\text{SD} = \frac{5}{\sqrt{40}} = 0.790\ldots$ **[1 mark]**

Margin of error $= 1.96 \times 0.790\ldots = 1.549\ldots$ **[1 mark]**

95% confidence interval $= [41.45, 44.55]$ **[1 mark]**

[3 marks altogether]

Part C – Analysis questions

1 For the 94% confidence interval, $z = \text{invNormCDf}(0.97,0,1) = 1.880\ldots$ **[1 mark]**

Margin of error $= \frac{7.7 - 7.2}{2} = 0.2$ **[1 mark]**

Sampling SD $= 0.2 \div 1.880\ldots = 0.106\ldots$ **[1 mark]**

SD of daily amount $= 0.106\ldots \times \sqrt{500} = 2.377\ldots$ **[1 mark]**

SD of sample of 8 $= \frac{2.377\ldots}{\sqrt{8}} = 0.840\ldots$ **[1 mark]**

Mean $= \frac{7.2 + 7.7}{2} = 7.45$ **[1 mark]**

normCDf$(-\infty, 8, 7.45, 0.840\ldots) = 0.743\ldots$ **[1 mark]**

If you asked another 8 people, the probability that they would sleep less than an average of 8 hours a day is about 0.74 **[1 mark]**

[8 marks altogether]

2 a It means that the 95% confidence interval margin of error is 4% of \$380 = \$15.20 **[1 mark]**

b Total population = 23.4 million **[1 mark]**

Numbers interviewed:

New South Wales $= \frac{7.5}{23.4} \times 800 \approx 256$

Victoria $= \frac{5.8}{23.4} \times 800 \approx 198$

Queensland ≈ 161

Western Australia ≈ 85

South Australia ≈ 58

Tasmania ≈ 17

Australian Capital Territory ≈ 14

Northern Territory ≈ 10 **[4 marks]**

c Standard deviation of amount spent $= 15.2 \div 1.96 \times \sqrt{800} = \$219.35$ **[1 mark]**

95% margins of error:

New South Wales $= 1.96 \times \frac{219.34}{\sqrt{256}} \approx \$26.87$

Victoria $= 1.96 \times \frac{219.34}{\sqrt{198}} \approx \$30.55$

Queensland $\approx \$33.88$

Western Australia $\approx \$46.63$

South Australia $\approx \$56.45$

Tasmania $\approx \$104.27$

Australian Capital Territory $\approx \$114.90$

Northern Territory $\approx \$135.95$ **[4 marks]**

d The survey covered only the capital cities, and did not include Australians living in regional and rural areas, which are likely to have higher fees.

Telephone polls are taken on landlines to ensure the correct distribution, but many people now, particularly younger people, do not have landlines.

There is no guarantee that the person answering is an adult on the electoral roll.

There will also be recall and non-response bias.

While the national figures for those in the capital cities are likely to be accurate, the much lower accuracies for smaller state and territories means that individual state and territory results and comparisons are not very accurate. **[3 marks: 1 for each point]**

[14 marks altogether]